

Moritz Dück

Full-stack product engineer · end-to-end ownership · data-heavy interfaces

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PROFILE

Full-stack engineer with six-plus years building data-heavy products end-to-end — frontend, backend, and cloud. As CTO of a seven-person team I own the whole lifecycle: understanding the problem, designing the solution, building it, testing it, and running it in production. The work I am proudest of sits where those ends meet — storage formats shaped by the interface they have to feed, and interfaces made simpler by knowing what the system can actually do. ETH training and published human-computer-interaction research back the design side of that.

EXPERIENCE

Simunto · Zurich, Switzerland

Sep 2021 – Present

Chief Technology Officer (Nov 2024 – Present) · Full-Stack Software Engineer (Sep 2021 – Nov 2024)

- Shipped the Mobility Impact Analyser for Volkswagen's MOIA — embedding with their team from requirements through to an AWS-deployed full-stack app for spatial analysis and large-scale simulation. It became the core asset in MOIA's mobility-consulting portfolio, turning bespoke studies into repeatable, self-service ride-pooling analyses for cities.
- Engineered the geospatial interfaces and data backbone behind Simunto's products — interactive network editors handling up to 10 million links (Vue, TypeScript, MapLibre GL, deck.gl) on top of Java pipelines processing up to 200 GB of simulation output per run.
- Led the core architecture decision behind Tramola's network analyses — purpose-built binary-encoded data structures that stream straight into browser typed arrays, keeping millions of road links interactive without a general-purpose query engine in the path. The encoding is derived from the interface it feeds; fields no analysis needs never enter the format.
- As CTO of a seven-person team, own product and technical direction end-to-end across the portfolio — problem framing, design, implementation, testing, and delivery.

ThoughtWorks · Berlin, Germany

Jan 2020 – Sep 2021

Software Developer Consultant

- Embedded in a client team in the payments domain, collaborating with client engineers and stakeholders to onboard new payment methods and improve observability across a microservice architecture.
- Operated 8 Java microservices with 5 colleagues on a Kubernetes cluster on AWS, processing up to 1,000 requests per second.
- Refactored the shopping-cart component to raise its availability SLO from 99.0% to 99.5% on a highly distributed e-commerce platform (Scala, functional).

- Worked in an engineering culture where test-driven development, pair programming, trunk-based development, and continuous delivery were the default – practices I have carried into every team since.

EDUCATION

ETH Zürich

Sep 2021 – Jul 2024

M.Sc. Computer Science – Major in Machine Intelligence, Minor in Data Management Systems

Otto-Friedrich University Bamberg

Sep 2016 – Sep 2019

B.Sc. Applied Computer Science – graduated with distinction (GPA 4.0/4.0)

PUBLICATIONS AND SELECTED WORK

- **Finding Needles in Document Haystacks: Augmenting Serendipitous Claim Retrieval Workflows** – ACM CHI 2025. Peer-reviewed HCI publication based on master thesis: an interactive system to support claim retrieval across large document collections. doi.org/10.1145/3706598.3713715
- **Neighborhood Traces (Interactive Explainable for CNNs)** – IEEE VISxAI 2023. First-authored interactive “explorable explainer”: three novel d3.js visualizations of how a CNN reshapes its latent space, built as a scrollytelling web experience (Vue.js + Flask). With H. Strobel (IBM Research) and M. El-Assady (ETH). moritzdueck.com/neighborhood-traces

SKILLS

BACKEND / DATA	Java, Spring Boot, Scala, Python, PostgreSQL, MongoDB, Neo4j, Cassandra
FRONTEND / VIZ	Vue, TypeScript, Angular, React, Redux, d3.js, deck.gl, MapLibre GL
CLOUD / DEVOPS	AWS (incl. CDK), Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Git
PRACTICES	Test-driven development, Cypress, continuous delivery, pair programming, agile
LANGUAGES	German (native), English (proficient), French (conversational)